

Data Communications & Ethics Final Project Report

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Course	DATA 201: Data Communications & Ethics
Assignment	Final Project Report

1. Introduction

1.1. Background & Motivation

The Data Communications and Ethics Final Project transforms a personal three-year academic journey at Parami University into a structured, data-driven narrative. The central question is: **How has my course selection, workload, and extracurricular involvement evolved over time and what does that evolution reveal about my growth as a data science student and leader?**

The story is meaningful because it documents a deliberate transformation. In Year 1, the academic load was broad, which is 15 humanities/social credits against just 3 technical credits per semester, reflecting an exploratory undergraduate mindset. By Year 3, Fall 2025, that ratio had fully inverted, a 20-credit semester composed entirely of technical and data-related courses. This is not gradual drift, but it rather is an accelerating, intentional specialisation toward data science.

A parallel narrative strand explores personal balance through a hobbies dataset of 30 curated favourites spanning music, film, television, and visual art, the creative outlets that sustained high performance under a demanding workload.

1.2. Dataset Design

Three original datasets were created for this project:

- ***khant_growth_journey.csv*** — The primary academic dataset. Contains one row per semester (8 semesters) with columns for Academic Year, Semester label, Tech/Data Credits, Humanities/Social Credits, Semester GPA, Estimated Weekly Club Hours, and a self-assessed Leadership Impact Score (1–5 scale).
- ***Khant_Journey.csv*** — A complementary raw journey file capturing broader semester metadata, including the 'IP' (In Progress) placeholder for the Spring 2026 GPA to preserve data truthfulness.

- **khant_hobbies.csv** — The secondary personal dataset. Contains 30 rows, one per favourite item, with columns for Category (Song / Movie / Series / Art), Title, Artist/Creator, Year, and a personal Rating out of 10.

All variables are clearly defined. No sensitive personal data, real third-party names beyond public artists and titles, or private institutional records are included.

1.3. Summary of Academic Dataset

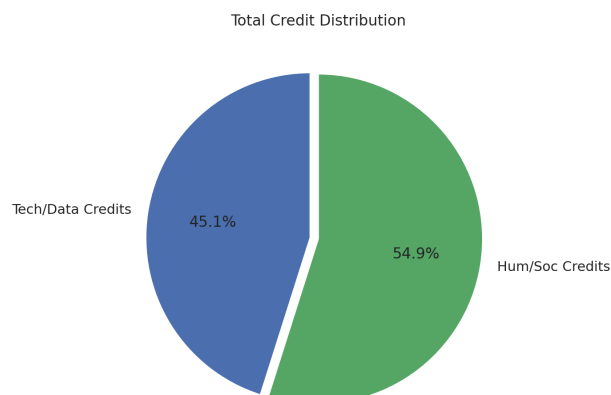
Semester	Credits	Tech Cr.	Hum Cr.	GPA	Club Hrs/wk	Leadership Role
Fall 2023 (Year 1)	18	3	15	3.87	10	CEDI
Spring 2024 (Year 1)	18	6	12	3.66	10	—
Fall 2024 (Year 2)	16	3	13	3.44	8	CIIF
Spring 2025 (Year 2)	14	2	12	3.80	8	—
Fall 2025 (Year 3)	20	20	0	3.53	4	—
Spring 2026 (Year 3)	16	12	4	IP	4	Borderless Futures

IP = In Progress. Spring 2026 GPA excluded from all visualisations to preserve truthfulness.

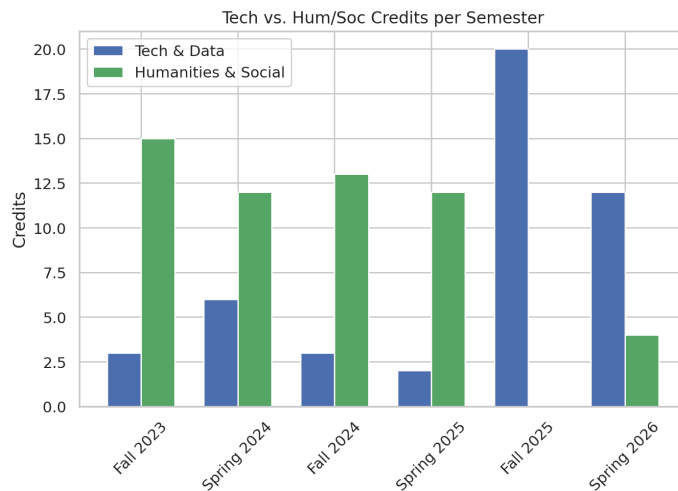
The data reveals a clear three-phase structure. Year 1 (Fall 2023 – Spring 2024): broad curriculum (15 humanities credits), high club involvement (10 hrs/week), Leadership Impact Score of 3, led by CEDI. Year 2 (Fall 2024 – Spring 2025): transitional phase, attempted credits decrease (16–14), GPA recovers to 3.80, leadership deepens with CIIF. Year 3 (Fall 2025 – Spring 2026): full technical immersion (20 tech credits in Fall 2025), club hours halved to 4/week, Leadership Impact Score peaks at 5.

2. Key Findings & Exploratory Data Analysis

2.1. The Academic Focus



Visualization 1 (Pie Chart) shows the aggregate credit split across all six semesters. Tech/Data credits total 46, humanities/social credits total 56, an overall near-even split.

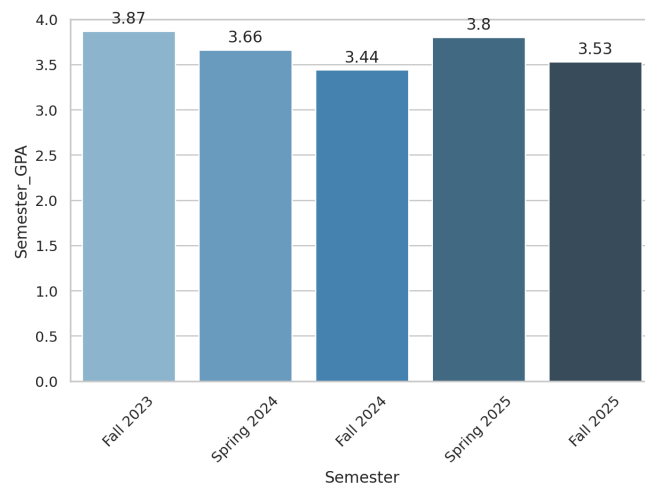


Visualization 2 (Grouped Bar Chart) exposes the true story. In Year 1, each semester carries 3–6 tech credits against 12–15 humanities credits. By Year 3 Fall 2025, the balance completely inverts: 20 tech credits, zero humanities. The pivot point is Fall 2024 (Year 2), where the first deliberate reduction in humanities credits occurred..

Key Finding 1

Humanities credits dominated Year 1 (15 per semester) and Year 2 (12–13 per semester). By Year 3 Fall 2025, humanities dropped to zero for the first time, and tech credits reached their maximum of 20. The overall aggregate ratio masks this complete inversion, the bar chart is essential for the true narrative.

2.2 Workload vs. Academic Performance

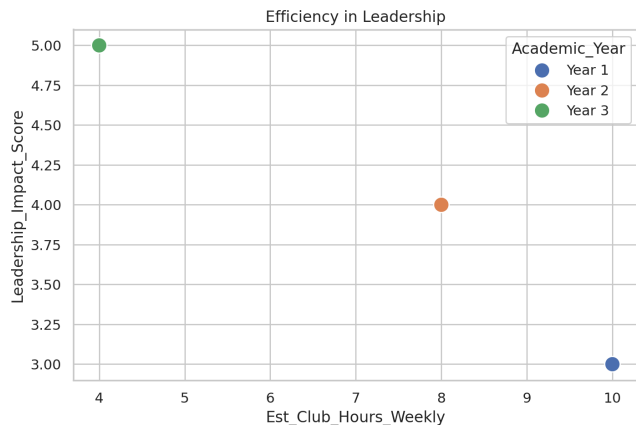


Visualization 3 (Bar Chart) shows a non-linear GPA trajectory across five recorded semesters: $3.87 \rightarrow 3.66 \rightarrow 3.44 \rightarrow 3.80 \rightarrow 3.53$. The lowest GPA occurred in Fall 2024 (Year 2, 3.44), coinciding with the transition semester where CIIF leadership began and course difficulty increased. Spring 2025 showed a strong recovery to 3.80, the highest recorded GPA. Fall 2025 dipped to 3.53 despite the maximum 20-credit load.

Key Finding 2

GPA was not monotonically increasing. The dip to 3.44 in the transition year (Fall 2024) followed by a rebound to 3.80 (Spring 2025) is the most important data point. It shows that the specialisation pivot initially disrupted performance before stabilising, which is a realistic pattern that an upward-only trend would have hidden.

2.3 Leadership Efficiency



Visualisation 4 (Scatter Plot) plots Estimated Weekly Club Hours against Leadership Impact Score, coloured by Academic Year. Year 1: 10 hours/week, score of 3. Year 2: 8 hours/week, score of 4. Year 3: 4 hours/week, score of 5. The inverse relationship is consistent across all three years.

Key Finding 3

Club hours halved from Year 1 (10 hrs/week) to Year 3 (4 hrs/week), yet Leadership Impact Score rose from 3 to 5. The shift from broad involvement (Debate, Datalliance, Environmental club) to a single high-stakes initiative (Debate Club) produced greater measurable impact with significantly less time investment.

2.4 Personal Balance

The secondary dataset tracks 30 personal favourites. Song is the largest category with 16 entries, followed by Movie (6), Series (5), and Art (3). The overall average rating is 9.03 out of 10.

Category	# Items	Notable Titles / Artists	Avg Rating /10
Song	16	Taylor Swift, Adele, Sia, Shawn Mendes, Charlie Puth, Alan Walker	8.94
Movie	6	Inception, Interstellar, HP series, Red Cliff, Lord of the Rings	9.67
Series	5	Game of Thrones, Breaking Bad, Stranger Things, The Crown, HotD	8.60
Art	3	Starry Night, The Great Wave, Mona Lisa	9.00

Key Finding 4

Movies achieve the highest average rating (9.67/10) despite having only 6 entries, every film on the list is exceptional (Inception, Interstellar, Lord of the Rings, Harry Potter). Songs dominated by volume (16 entries, 8.94 avg) because music integrates seamlessly into study and coding sessions without displacing focus time. The minimum rating across all 30 items is 8/10, confirming that only top-tier content earns a place on the list.

3. Data Decision Making

3.1. Continue Technical Specialisation in Senior Year and Postgraduate Study

The data from the credit funnel (Visualizations 1 and 2) shows that the change to full technical involvement in Year 3 was planned and done. A 3.80 GPA in the spring of 2025 was the best ever. This was after a year of fewer arts classes.

Recommendation: I should get a master's degree in Data Science or a technical area similar to it. The results show that focused, similar coursework is linked to better academic success for me.

Limitation: The high GPA in the spring of 2025 might not just be due to specialization, but also to a lower total class load (only 14 credits, the lowest on record).

3.2. Prioritise One High-Responsibility Leadership Role

When there are fewer club hours and one main leadership role, the scatter plot (Visualization 4) constantly shows that the effect scores are higher.

Recommendation: I should not feel the need to join a lot of clubs or groups. When it comes to leadership growth, one well-chosen, high-stakes project (like Debate Club) works better than a lot of low-stakes activity.

Limitation: One problem is that the Leadership Impact Score is self-rated on a scale from 1 to 5. The upward trend isn't just due to objective external effects; it's also due to rising trust and job seniority. It would be stronger if it were supported by outside evidence, like peer review or organizational effects.

3.3. Maintain Music as a Daily Study Companion

The hobbies data has 16 song titles, which is by far the most and has an average score of 8.94/10. There are a lot of songs by artists like Alan Walker (EDM), Taylor Swift (Pop), and Adele (Pop/Soul).

Recommendation: As a conscious way to get work done, I should keep putting together a personal study playlist. Songs, unlike movies or TV shows, don't need to be watched, so they can be played right during technical work sessions without stopping the flow.

Limitation: One problem is that the hobbies dataset only records stated interests and not real listening time or measured effects on productivity. From this data alone, we can't say for sure if music actually makes people better at school.

4. Ethics & Responsibility Disclosures

4.1 Privacy & Data Scope

- All data is self-generated. Academic metrics are personal records, not extracted from institutional systems.
- Active Clubs column lists organisation names only. No individual members, instructors, or third-party collaborators within CEDI, CIIF, Debate, Datalliance, or Borderless Futures are named or identified.
- The hobbies dataset references only publicly available works and artists. No private viewing history, streaming data, or platform data was used.
- No official transcripts, grade reports, or institutional communications are reproduced anywhere in the dataset or dashboard.

4.2 Bias & Limitations

- **Memory Bias:** Est_Club_Hours_Weekly numbers are estimates based on the past; they are not kept logs. Years 1–2 numbers, in particular, are hard to remember. They are not exact lengths, so they should only be used as rough guides.
- **Small Dataset:** The academic dataset only has six rows. There are no valid inferential data. All of the designs are only descriptive and experimental.
- **Subjective Scoring:** The Leadership Impact Score (1–5) is a personal rating that the person giving it gives themselves. It hasn't been checked by anyone outside the company, so it can be biased in its own favor and lose its accuracy over time.
- **Progress GPA:** GPA for the current semester, spring 2026, is "IP," which stands for "In Progress." To avoid showing the wrong partial-semester GPA, it is kept as NaN in the main dataset and left out of all visualizations.
- **Confounding Variables:** Changes in course difficulty, teacher grading standards, and personal factors that aren't included in the dataset may also explain changes in GPA as much as the measured variables.

4.3 Visualisation Justification & Risk of Misinterpretation

- **Pie Chart (Overall Credit Split):** It is convenient for a static, two-way view of the total amount of points. The risk is that the sum hides the reversal of time. It can be lessened by pairing straight with the grouped bar chart.
- **Grouped Bar Chart (Semesterly Credits):** Discrete bars correctly show semester blocks that don't run into each other in a grouped bar chart showing semester credits.
- **Bar Chart (GPA Trend):** A line chart would give the wrong impression of smooth, ongoing change between different time eras.
- **Scatter Plot (Leadership Efficiency):** The y-axis is set to 0.0 and can't go above 4.0 to keep the proportional accuracy. The GPA number is written on each bar so that the heights can't be misunderstood. As we can see, the Leadership Efficiency Scatter Plot displays the

coordinate-level link between two continuous factors over three academic years. There is no regression line because there are only 6 points, two per year, so it would not make sense analytically or physically to fit a line.

4.4 Responsible Decision Statement

All three choices in Section 3 are based on trends that have been seen in the data set and come with clear caveats. None of these findings are shown to be true for everyone. The academic dataset shows six semesters of one student's time at one school, and the hobbies dataset shows that person's carefully chosen interests. People who read or watch the dashboard are told to use the dynamic filters to look at the data from their own points of view instead of taking any one story frame at face value.